In [1]:
```
import numpy as np
import pandas as pd
```

In [2]:
```
launch_years = np.array([
    1969, 1977, 1996, 2008, 2009,
    2011, 2004, 2006, 2023, 2024
])

print(launch_years)
```

```
[1969 1977 1996 2008 2009 2011 2004 2006 2023 2024]
```

In [3]:
```
print("Dimensions:", launch_years.ndim)
print("Shape:", launch_years.shape)
print("Size:", launch_years.size)
```

```
Dimensions: 1
Shape: (10,)
Size: 10
```

In [4]:
```
print("Minimum:", np.min(launch_years))
print("Maximum:", np.max(launch_years))
print("Total:", np.sum(launch_years))
print("Average:", np.mean(launch_years))
```

```
Minimum: 1969
Maximum: 2024
Total: 20027
Average: 2002.7
```

In [5]:
```
print(launch_years.min())
print(launch_years.max())
print(launch_years.sum())
print(launch_years.mean())
```

```
1969
2024
20027
2002.7
```

In [6]:
```
mission_data = np.array([
    [1969, 8],
    [1977, 17520],
    [1996, 300],
    [2008, 312],
    [2009, 3400],
    [2011, 5000],
    [2004, 3900],
    [2006, 7000],
    [2023, 14],
    [2024, 2400]
])

print(mission_data)

print("Dimensions:", mission_data.ndim)
```

```
print("Shape:", mission_data.shape)
print("Size:", mission_data.size)
```

```
[[ 1969     8]
 [ 1977 17520]
 [ 1996   300]
 [ 2008   312]
 [ 2009  3400]
 [ 2011  5000]
 [ 2004  3900]
 [ 2006  7000]
 [ 2023    14]
 [ 2024  2400]]
Dimensions: 2
Shape: (10, 2)
Size: 20
```

In [7]:
```
print(np.min(mission_data, axis=1))
```

```
[   8 1977  300  312 2009 2011 2004 2006   14 2024]
```

In [8]:
```
df = pd.read_csv("space_missions.csv")

print(df.head())
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
```

```
  Mission_ID     Mission_Name Agency              Target  Launch_Year  \
0       M001        Apollo 11   NASA               Moon         1969
1       M002        Voyager 1   NASA  Interstellar Space         1977
2       M003  Mars Pathfinder   NASA                Mars         1996
3       M004    Chandrayaan-1   ISRO               Moon         2008
4       M005           Kepler   NASA          Exoplanets         2009

       Rocket  Duration_Days  Distance_Million_KM     Status  \
0    Saturn V              8         3.840000e-01  Completed
1  Titan IIIE          17520         2.400000e+04     Active
2   Delta II             300         7.800000e+01  Completed
3    PSLV-XL             312         3.840000e-01  Completed
4   Delta II            3400         1.500000e+08  Completed

               Objective
0          Lunar landing
1  Deep space exploration
2     Mars surface study
3          Lunar mapping
4       Planet discovery
Shape: (10, 10)
Columns: ['Mission_ID', 'Mission_Name', 'Agency', 'Target', 'Launch_Year', 'Rocket',
'Duration_Days', 'Distance_Million_KM', 'Status', 'Objective']
```

In [10]:
```
df.head()
```

Out[10]:

	Mission_ID	Mission_Name	Agency	Target	Launch_Year	Rocket	Duration_Days	Di
0	M001	Apollo 11	NASA	Moon	1969	Saturn V	8	
1	M002	Voyager 1	NASA	Interstellar Space	1977	Titan IIIE	17520	
2	M003	Mars Pathfinder	NASA	Mars	1996	Delta II	300	
3	M004	Chandrayaan-1	ISRO	Moon	2008	PSLV-XL	312	
4	M005	Kepler	NASA	Exoplanets	2009	Delta II	3400	

In [12]:
```
df.tail()
```

Out[12]:

	Mission_ID	Mission_Name	Agency	Target	Launch_Year	Rocket	Duration_Days	Distar
5	M006	Curiosity	NASA	Mars	2011	Atlas V	5000	
6	M007	Rosetta	ESA	Comet 67P	2004	Ariane 5G+	3900	
7	M008	New Horizons	NASA	Pluto	2006	Atlas V	7000	
8	M009	Chandrayaan-3	ISRO	Moon	2023	LVM3	14	
9	M010	Europa Clipper	NASA	Europa	2024	Falcon Heavy	2400	

In [14]:
```
df.shape
df.columns
```

Out[14]:
```
Index(['Mission_ID', 'Mission_Name', 'Agency', 'Target', 'Launch_Year',
       'Rocket', 'Duration_Days', 'Distance_Million_KM', 'Status',
       'Objective'],
      dtype='str')
```

In [16]:
```
df.shape
```

Out[16]:
```
(10, 10)
```

In [18]:
```
df.describe()
```

Out[18]:

	Launch_Year	Duration_Days	Distance_Million_KM
count	10.000000	10.000000	1.000000e+01
mean	2002.700000	3985.400000	1.500312e+07
std	17.814164	5321.172306	4.743307e+07
min	1969.000000	8.000000	3.840000e-01
25%	1998.000000	303.000000	1.978800e+01
50%	2007.000000	2900.000000	2.890000e+02
75%	2010.500000	4725.000000	4.582000e+03
max	2024.000000	17520.000000	1.500000e+08

In [20]:
```
df.isnull().sum()
```

Out[20]:
```
Mission_ID             0
Mission_Name           0
Agency                 0
Target                 0
Launch_Year            0
Rocket                 0
Duration_Days          0
Distance_Million_KM    0
Status                 0
Objective              0
dtype: int64
```

In [22]:
```
df.duplicated().sum()
```

Out[22]:
```
np.int64(0)
```

In [24]:
```
df["Duration_Years"] = df["Duration_Days"] / 365.25

print(df[["Mission_Name", "Duration_Days", "Duration_Years"]])
```

```
     Mission_Name  Duration_Days  Duration_Years
0        Apollo 11              8        0.021903
1        Voyager 1          17520       47.967146
2  Mars Pathfinder            300        0.821355
3    Chandrayaan-1            312        0.854209
4           Kepler           3400        9.308693
5        Curiosity           5000       13.689254
6          Rosetta           3900       10.677618
7     New Horizons           7000       19.164956
8    Chandrayaan-3             14        0.038330
9   Europa Clipper           2400        6.570842
```

In [26]:
```
print("Minimum duration:", df["Duration_Days"].min())
print("Maximum duration:", df["Duration_Days"].max())
print("Average duration:", df["Duration_Days"].mean())
print("Total duration:", df["Duration_Days"].sum())
```

```
Minimum duration: 8
Maximum duration: 17520
Average duration: 3985.4
Total duration: 39854
```

In [28]:
```
print(df[df["Agency"] == "NASA"])
```

```
  Mission_ID     Mission_Name Agency              Target  Launch_Year  \
0       M001        Apollo 11   NASA                Moon         1969
1       M002        Voyager 1   NASA  Interstellar Space         1977
2       M003  Mars Pathfinder   NASA                Mars         1996
4       M005           Kepler   NASA          Exoplanets         2009
5       M006        Curiosity   NASA                Mars         2011
7       M008     New Horizons   NASA               Pluto         2006
9       M010   Europa Clipper   NASA              Europa         2024

         Rocket  Duration_Days  Distance_Million_KM      Status  \
0      Saturn V              8         3.840000e-01   Completed
1    Titan IIIE          17520         2.400000e+04      Active
2      Delta II            300         7.800000e+01   Completed
4      Delta II           3400         1.500000e+08   Completed
5       Atlas V           5000         7.800000e+01      Active
7       Atlas V           7000         5.900000e+03      Active
9  Falcon Heavy           2400         6.280000e+02  In Transit

                        Objective  Duration_Years
0                   Lunar landing        0.021903
1           Deep space exploration       47.967146
2               Mars surface study        0.821355
4                 Planet discovery        9.308693
5                Mars habitability       13.689254
7               Outer solar system       19.164956
9  Search for habitable conditions        6.570842
```

In [30]:
```
print(df[(df["Agency"] == "NASA") & (df["Target"] == "Mars")])
```

```
  Mission_ID     Mission_Name Agency Target  Launch_Year    Rocket  \
2       M003  Mars Pathfinder   NASA   Mars         1996  Delta II
5       M006        Curiosity   NASA   Mars         2011   Atlas V

   Duration_Days  Distance_Million_KM     Status           Objective  \
2            300                 78.0  Completed  Mars surface study
5           5000                 78.0     Active   Mars habitability

   Duration_Years
2        0.821355
5       13.689254
```

In [33]:
```
print(df.groupby("Agency").size())
```

```
Agency
ESA     1
ISRO    2
NASA    7
dtype: int64
```

In [35]:
```
print(df.sort_values("Launch_Year", ascending=False))
```

	Mission_ID	Mission_Name	Agency	Target	Launch_Year	\
9	M010	Europa Clipper	NASA	Europa	2024	
8	M009	Chandrayaan-3	ISRO	Moon	2023	
5	M006	Curiosity	NASA	Mars	2011	
4	M005	Kepler	NASA	Exoplanets	2009	
3	M004	Chandrayaan-1	ISRO	Moon	2008	
7	M008	New Horizons	NASA	Pluto	2006	
6	M007	Rosetta	ESA	Comet 67P	2004	
2	M003	Mars Pathfinder	NASA	Mars	1996	
1	M002	Voyager 1	NASA	Interstellar Space	1977	
0	M001	Apollo 11	NASA	Moon	1969	

	Rocket	Duration_Days	Distance_Million_KM	Status	\
9	Falcon Heavy	2400	6.280000e+02	In Transit	
8	LVM3	14	3.840000e-01	Completed	
5	Atlas V	5000	7.800000e+01	Active	
4	Delta II	3400	1.500000e+08	Completed	
3	PSLV-XL	312	3.840000e-01	Completed	
7	Atlas V	7000	5.900000e+03	Active	
6	Ariane 5G+	3900	5.000000e+02	Completed	
2	Delta II	300	7.800000e+01	Completed	
1	Titan IIIE	17520	2.400000e+04	Active	
0	Saturn V	8	3.840000e-01	Completed	

	Objective	Duration_Years
9	Search for habitable conditions	6.570842
8	Lunar soft landing	0.038330
5	Mars habitability	13.689254
4	Planet discovery	9.308693
3	Lunar mapping	0.854209
7	Outer solar system	19.164956
6	Comet exploration	10.677618
2	Mars surface study	0.821355
1	Deep space exploration	47.967146
0	Lunar landing	0.021903

In [37]:
```
df.isnull().sum()
```

Out[37]:
```
Mission_ID            0
Mission_Name          0
Agency                0
Target                0
Launch_Year           0
Rocket                0
Duration_Days         0
Distance_Million_KM   0
Status                0
Objective             0
Duration_Years        0
dtype: int64
```

In [38]:
```
df.describe(include = "all")
```

Out[38]:

	Mission_ID	Mission_Name	Agency	Target	Launch_Year	Rocket	Duration_Days
count	10	10	10	10	10.000000	10	10.000000
unique	10	10	3	7	NaN	8	NaN
top	M001	Apollo 11	NASA	Moon	NaN	Delta II	NaN
freq	1	1	7	3	NaN	2	NaN
mean	NaN	NaN	NaN	NaN	2002.700000	NaN	3985.400000
std	NaN	NaN	NaN	NaN	17.814164	NaN	5321.172306
min	NaN	NaN	NaN	NaN	1969.000000	NaN	8.000000
25%	NaN	NaN	NaN	NaN	1998.000000	NaN	303.000000
50%	NaN	NaN	NaN	NaN	2007.000000	NaN	2900.000000
75%	NaN	NaN	NaN	NaN	2010.500000	NaN	4725.000000
max	NaN	NaN	NaN	NaN	2024.000000	NaN	17520.000000